

第1章

4. 解: (1) ③ (2) ④ (3) ⑤ (4) ⑦ (5) ⑥

5. 解: (1) 正确 (5) 正确
(2) 不正确 (6) 正确
(3) 正确 (7) 正确
(4) 正确 (8) 不正确.

16. 解: (1) $UA = \{1, 2, 3, \emptyset\}$

(2) $UA = \emptyset$

38. 解: (1) $\sim A$ 对应的 0-1 串是 01001101.

(2) 依题知: $\forall k \in \{1, 2, \dots, n\}$

$$a_k = i_k \vee j_k = \begin{cases} 1, & i_k \neq 0 \text{ 或 } j_k \neq 0 \\ 0, & i_k = 0 \text{ 且 } j_k = 0. \end{cases}$$

$$b_k = i_k \& j_k = \begin{cases} 1, & i_k = 1 \text{ 且 } j_k = 1 \\ 0, & i_k = 0 \text{ 或 } j_k = 0. \end{cases}$$

$$c_k = \begin{cases} 1, & i_k = 1 \text{ 且 } j_k = 0 \\ 0, & i_k = 0 \text{ 或 } (i_k = 1 \text{ 且 } j_k = 1) \end{cases}$$

$$d_k = i_k \wedge j_k = \begin{cases} 1, & i_k \neq j_k \\ 0, & i_k = j_k. \end{cases}$$

40. 解: (1) $P(A) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\{\emptyset\}, \{\{\emptyset\}\}\}\}$

(2) $UA = \{\emptyset, \{\emptyset\}\}$

$P(UA) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\{\emptyset, \{\emptyset\}\}\}$

(3) $UP(A) = A = \{\{\emptyset\}, \{\{\emptyset\}\}\}$

45. 解: 设全集 $E = \{1, 2, \dots, 300\}$, 能被 4 整除的整数

集合为 $M_k, M_k \subseteq E$. 依题知:

$$|M_3| = 100, |M_5| = 60, |M_7| = 42$$

$$(1) |M_3 \cap M_5 \cap M_7| = \left\lfloor \frac{|E|}{\text{lcd}(3, 5, 7)} \right\rfloor = \left\lfloor \frac{300}{105} \right\rfloor = 2.$$

$$(2) |\overline{M_3} \cap \overline{M_5} \cap \overline{M_7}| = |E| - (|M_3| + |M_5| + |M_7|) + (|M_3 \cap M_5| + |M_3 \cap M_7| + |M_5 \cap M_7|) - |M_3 \cap M_5 \cap M_7|$$

$$= 300 - 202 + 20 + 14 + 8 - 2 = 138$$

$$(3) |M_3 \cap \overline{M_5} \cap \overline{M_7}| = |M_3| - |M_3 \cap M_5| - |M_3 \cap M_7| + |M_3 \cap M_5 \cap M_7|$$

$$= 100 - 20 - 14 + 2 = 68.$$

$$(4) |M_3 \cup M_5 - M_7|$$

$$= |M_3 \cup M_5| - |M_3 \cap M_7| - |M_5 \cap M_7| + |M_3 \cap M_5 \cap M_7|$$

$$= 160 - 20 - 14 - 8 + 2 = 120.$$

$$(5) |M_3 \cap \overline{M_5} \cap \overline{M_7}| = 68$$

$$|\overline{M_3} \cap M_5 \cap \overline{M_7}| = 60 - 20 - 8 + 2 = 34$$

$$|\overline{M_3} \cap \overline{M_5} \cap M_7| = 42 - 8 - 14 + 2 = 22$$

满足题意的整数共 $68 + 34 + 22 = 124$ 个.

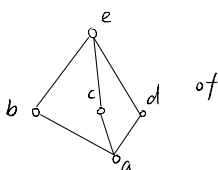
第二章

36. (2) 解: 依题知:

$$A \times A / R = \{ [\langle 2, 1 \rangle]_R, [\langle 3, 1 \rangle]_R, [\langle 4, 1 \rangle]_R, [\langle 1, 2 \rangle]_R, [\langle 1, 3 \rangle]_R, [\langle 4, 4 \rangle]_R, [\langle 1, 1 \rangle]_R \}$$

$$= \{ \{ \langle 2, 1 \rangle, \langle 3, 2 \rangle, \langle 4, 3 \rangle \}, \{ \langle 3, 1 \rangle, \langle 4, 2 \rangle \}, \{ \langle 4, 1 \rangle \}, \{ \langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 3, 4 \rangle \}, \{ \langle 1, 3 \rangle, \langle 2, 4 \rangle \}, \{ \langle 1, 4 \rangle \}, \{ \langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle, \langle 4, 4 \rangle \} \}$$

46. 解: (1) 哈斯图如下:



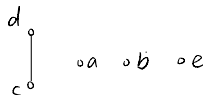
极大元: e, f

极小元: a, f

最大元: 无

最小元: 无

(2) 哈斯图如下:



极大元: d, a, b, e

极小元: c, a, b, e

最大元: 无

最小元: 无

49. 解: (1) 证: 依题知: R 是自反、反对称、传递的.

$$\textcircled{1} \forall x \in A, \langle x, x \rangle \in R \Leftrightarrow xRx$$

$$\Rightarrow xSx \Rightarrow S \text{ 是自反的.}$$

$$\textcircled{2} \forall x, y \in A, xRy \wedge yRx \Rightarrow x=y.$$

$$\forall u, v \in A, uSv \wedge vSu \Leftrightarrow vRu \wedge uRv$$

$$\Rightarrow u=v. \Rightarrow S \text{ 是反对称的.}$$

$$\textcircled{3} \forall x, y, z \in A, xSy \wedge ySz \Leftrightarrow yRx \wedge zRy$$

$$\Rightarrow zRx \Leftrightarrow xSz \Rightarrow S \text{ 是传递的.}$$

综上所述, S 是 A 上的偏序关系. ■

(2) S 是大于等于关系;

S 是覆盖关系.

(3) $\langle A, R \rangle$ 中的极大(小)元是

$\langle A, S \rangle$ 中的极小(大)元.

$\langle A, R \rangle$ 中的最大(小)元是

$\langle A, S \rangle$ 中的最小(大)元. (若有存在).

10. 解: (1) $\langle 2, b \rangle, \langle 3, 1 \rangle \in R \cup S$ 答案中有 $\langle 0, 0 \rangle$?

$$(2) \langle 2, 1 \rangle \in R \cup S$$

(3) 没有满足条件的有序对

16. 解: $\textcircled{1} R_1 \circ R_2 = \{ \langle a, d \rangle, \langle a, c \rangle \}$

$$\textcircled{2} R_2 \circ R_1 = \{ \langle c, d \rangle \}$$

$$\textcircled{3} R_1^2 = \{ \langle a, a \rangle, \langle a, b \rangle, \langle a, d \rangle \}$$

$$\textcircled{4} R_2^2 = \{ \langle b, b \rangle, \langle c, c \rangle, \langle c, d \rangle \}$$

$$R_3 = R_2 \circ R_1^2 = \{ \langle b, c \rangle, \langle c, b \rangle, \langle b, d \rangle \}$$

26. 解: $R = \{ \langle 1, 5 \rangle, \langle 2, 5 \rangle, \langle 1, 3 \rangle, \langle 4, 5 \rangle, \langle 3, 3 \rangle \}$

$$(1) R^2 = \{ \langle 1, 3 \rangle, \langle 3, 3 \rangle \}$$

$$R^3 = R^2 \circ R = \{ \langle 1, 3 \rangle, \langle 3, 3 \rangle \}$$

$$(2) r(R) = R \cup I_A = \{ \langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle, \langle 4, 4 \rangle, \langle 5, 5 \rangle, \langle 6, 6 \rangle, \langle 1, 5 \rangle, \langle 2, 5 \rangle, \langle 4, 5 \rangle, \langle 1, 3 \rangle \}$$

$$s(R) = R \cup R^{-1} = \{ \langle 1, 5 \rangle, \langle 5, 1 \rangle, \langle 2, 5 \rangle, \langle 5, 2 \rangle, \langle 1, 3 \rangle, \langle 3, 1 \rangle, \langle 4, 5 \rangle, \langle 5, 4 \rangle, \langle 3, 3 \rangle \}$$

$$t(R) = R \cup R^2 \cup R^3 \cup \dots$$

$$\text{由 (1) 知: } R^2 = R^3,$$

$$\Rightarrow t(R) = R \cup R^2 = \{ \langle 1, 5 \rangle, \langle 2, 5 \rangle, \langle 1, 3 \rangle, \langle 4, 5 \rangle, \langle 3, 3 \rangle \}.$$

31. 解: (1) 依题知:

$$R = \{ \langle x, y \rangle \mid x, y \text{ 在同一个等价类中}, x, y \in A \}$$

$$\Rightarrow R = \{ \langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle, \langle 4, 4 \rangle, \langle 2, 3 \rangle, \langle 3, 2 \rangle, \langle 2, 4 \rangle, \langle 4, 2 \rangle, \langle 3, 4 \rangle, \langle 4, 3 \rangle \}.$$

(2) R 是等价关系 $\Rightarrow R$ 是对称的

$$\Rightarrow R = R^{-1} \Rightarrow R \circ R^{-1} = R$$

(3) R 是等价关系 $\Rightarrow R$ 是传递的

$$\Rightarrow t(R) = R$$

36. 解: (1) 证: $\textcircled{1} \forall \langle x, y \rangle \in A \times A, x+y = x+y \quad \textcircled{2} \forall \langle u, v \rangle, \langle x, y \rangle, \langle \langle x, y \rangle, \langle a, b \rangle \rangle \in R$

$$\Rightarrow \langle \langle x, y \rangle, \langle x, y \rangle \rangle \in R,$$

$$\Rightarrow u+y = x+v, x+b = a+y$$

$$\Rightarrow R \text{ 是自反的}$$

$$\Rightarrow u+y+x+b = x+v+a+y$$

$$\textcircled{2} \forall \langle u, v \rangle, \langle x, y \rangle \in R$$

$$\Rightarrow u+b = a+v.$$

$$\Rightarrow u+y = x+v$$

$$\Rightarrow \langle \langle u, v \rangle, \langle a, b \rangle \rangle \in R.$$

$$\Rightarrow x+v = u+y$$

$$\Rightarrow R \text{ 是传递的.}$$

$$\Rightarrow \langle \langle x, y \rangle, \langle u, v \rangle \rangle \in R.$$

$$\Rightarrow R \text{ 是对称的.}$$

综上所述, R 是等价关系. ■